

Human Review: radial-twist sum-coordinate counterexample

Original automatic proof: `solution_packet.pdf`

Checked date: 10 June 2026

Status: Manually checked.

Verdict: Appears correct.

Human Comments

The solution appears to be correct. The central idea is to encode the information of a vector x into two coordinates, with the split depending on the radius $\|x\|$. Concretely, the construction has the form

$$F(x) = (\cos(\theta(\|x\|))x, \sin(\theta(\|x\|))x).$$

Thus the full map $F = (F_1, F_2)$ still remembers x in the two-coordinate sum, but each individual coordinate alternately loses all information on carefully chosen radial shells.

The alternation is governed by the norm of x . On one sequence of radii, $\theta(\|x\|) = 0$, so the second coordinate vanishes:

$$F_2(x) = 0.$$

On another sequence of radii, $\theta(\|x\|) = \pi/2$, so the first coordinate vanishes:

$$F_1(x) = 0.$$

The full map remains a coarse Lipschitz embedding because the information is not destroyed; it is merely moved between the two coordinates as the radius changes.

Strength of the Result

The packet proves something stronger than what is needed for the stated problem. It does not merely show that there is no infinite-dimensional subspace on which one coordinate becomes a coarse Lipschitz embedding. Rather, for every nonzero linear subspace X_0 , neither

$$F_1|_{X_0} \quad \text{nor} \quad F_2|_{X_0}$$

is a coarse embedding.

Indeed, after choosing any unit vector $u \in X_0$, the points $R_k u$ and $-R_k u$ become arbitrarily far apart in X_0 , while one of the coordinate maps sends both to the same point. The same argument applies to the other coordinate along the alternating sequence of radii. Thus the failure already occurs on every one-dimensional subspace, not merely on infinite-dimensional ones.

Short Version

The proof works by a radial twist. The vector x is split between two coordinates according to $\|x\|$, and the active coordinate alternates at larger and larger scales. The pair $F = (F_1, F_2)$ is still a coarse Lipschitz embedding, but each separate coordinate collapses arbitrarily distant pairs in every nonzero subspace.

Literature and Follow-Up

We did not find this result stated in the literature during the present review. Since the surrounding problem is specific and the construction is quite natural once seen, it may be good to reach out to the author to verify whether the example is already known and to get his take on its significance.

Review Outcome

Archive this packet as an apparently correct negative solution to the coarse-sum coordinate problem, with the additional note that the construction is stronger than the stated obstruction: no coordinate map works even after restriction to a nonzero finite-dimensional subspace.